Fengxu Yang

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RESEARCH INTERESTS

My research primarily focuses on **LiDAR-centric perception and the security and privacy of autonomous driving systems**. Previously, I worked on wireless network coexistence, heterogeneous networking, and cross-technology communication to enable robust connectivity for embedded devices.

EDUCATION

ShanghaiTech University Sept. 2022 – July 2026
Ph.D. in Computer Science and Technology
Supervisor: Zhice Yang

ShanghaiTech University Sept. 2019 – July 2022
Master's studies in Information and Communication Engineering (transferred to the Ph.D. program)
Joint training with Shanghai Advanced Research Institute, Chinese Academy of Sciences (SARI)
Research conducted at SARI under the supervision of Jianming Wei

North University of China Sept. 2014 – July 2018
B.Eng. in Information Countermeasure Technology

EXPERIENCE

Shanghai Advanced Research Institute, Chinese Academy of Sciences Feb. 2019 – Sept. 2019
Research Assistant, LoRa Testbed Project

PUBLICATIONS

Conference Papers

1. **(SECON'26, CCF-B)** Jiaqi Zhou, **Fengxu Yang**, Yihui Yan, and Zhice Yang, "Enabling Seamless LoRa-to-Mobile Bridging via Wi-Fi Spectrum Scanning", IEEE International Conference on Sensing, Communication, and Networking, 2026.
2. **(RAID'25, CCF-B, CORE A)** **Fengxu Yang**, Paizhuo Chen, Yihui Yan, and Zhice Yang, "An In-model Spy in Edge Intelligence", The 28th International Symposium on Research in Attacks, Intrusions and Defenses, 2025.
3. **(SenSys'25, CCF-B, CORE A*)** **Fengxu Yang**, Zaizhou Yang, Jiaqi Zhou, and Zhice Yang, "LiDARMarker: Machine-friendly Road Markers for Smart Driving Systems", The 23rd ACM Conference on Embedded Networked Sensor Systems, 2025.
4. **(INFOCOM'25, CCF-A, CORE A*)** Si Liao, **Fengxu Yang**, Huangxun Chen, and Zhice Yang, "Enable Autonomous Backscatter in Everyday Devices", IEEE International Conference on Computer Communications, 2025.
5. **(IPSN'22, CCF-B, CORE A*)** **Fengxu Yang**, Pei Tian, Xiaoyuan Ma, Carlo Alberto Boano, Ye Liu, and Jianming Wei, "EMU: Increasing the Performance and Applicability of LoRa through Chirp Emulation, Snipping, and Multiplexing", The 21st ACM/IEEE International Conference on Information Processing in Sensor Networks, 2022.
6. **(EWSN'21, CORE A)** Pei Tian, Xiaoyuan Ma, Carlo Alberto Boano, Ye Liu, **Fengxu Yang**, Xin Tian, Dan Li, and Jianming Wei, "ChirpBox: An Infrastructure-Less LoRa Testbed", The International Conference on Embedded Wireless Systems and Networks, 2021.

Journal Articles

1. **(Sensors'21)** Qingjie Guo, **Fengxu Yang**, and Jianming Wei, "Experimental Evaluation of the Packet Reception Performance of LoRa", Sensors, 2021, 21(4): 1071.

Workshops, Demos, and Posters

1. **(SenSys'25)** Zaizhou Yang, Yihui Yan, **Fengxu Yang**, and Zhice Yang, "Demo Abstract: Using Fingerprint Scanner for On-Body Messaging", The 23rd ACM Conference on Embedded Networked Sensor Systems, 2025.

2. **(SenSys'25)** Yihong Hang, Hao Li, **Fengxu Yang**, Zhice Yang, and Huangxun Chen, "Poster Abstract: LLM-Piloted Visual Privacy Agent on Mobile Systems", The 23rd ACM Conference on Embedded Networked Sensor Systems, 2025.
3. **(EWSN'22)** **Fengxu Yang**, Pei Tian, Xiaoyuan Ma, Jianming Wei, and Carlo Alberto Boano, "Demo: Real-Time Decoding of LoRa Packets Without Prior Knowledge of their Spreading Factor", The International Conference on Embedded Wireless Systems and Networks, 2022.
4. **(DATA'21 (SenSys Workshop))** Pei Tian, **Fengxu Yang**, Xiaoyuan Ma, Carlo Alberto Boano, Xin Tian, Ye Liu, and Jianming Wei, "Dataset: Environmental Impact on the Long-Term Connectivity and Link Quality of an Outdoor LoRa Network", The 19th ACM Conference on Embedded Networked Sensor Systems, 2021.
5. **(EWSN'20)** Xiaoyuan Ma, Dan Li, **Fengxu Yang**, Carlo Alberto Boano, Pei Tian, and Jianming Wei, "Poster: Chirpbox – A Low-Cost LoRa Testbed Solution", The International Conference on Embedded Wireless Systems and Networks, 2020.

Patents

1. **Fengxu Yang**, Yu An, Bobo Zhou, Shaoyu Wang, and Huitong Fu, "Elderly Mobility Aid Monitoring Device Based on the BeiDou Satellite Navigation System", CN206479642U (granted).

HONORS & AWARDS

- ACM SIGBED Student Travel Grant, CPS-IoT Week (10/110), 2025
- SIST Outstanding Teaching Assistant Award, ShanghaiTech University, 2023
- Second Prize, Future Light – Future Network Technology Innovation Competition, 2022
- Best Paper Award Nominee, EWSN, 2021
- Best Poster Award, EWSN, 2020
- National Undergraduate Innovation and Entrepreneurship Training Program, 2017
- Third Prize, International Underwater Robot Competition, 2017
- First Prize, Shanxi Province Undergraduate Electronic Design Contest, 2016
- Third Prize, National College Student Smart Internet Innovation Competition, 2016
- Third Prize, International Underwater Robot Competition, 2016
- Third Prize, China Robot Competition, 2016
- First Prize, International Underwater Robot Competition, 2015
- Second Prize, National Undergraduate Electronics Design Contest, 2015
- Second Prize, Robot Competition for College Students in Five Provinces of North China, 2015

TEACHING EXPERIENCE

Teaching Assistant

Computer Networks (CS120), Fall 2022, Fall 2023, and Fall 2026
Wireless and Mobile Systems (CS222), Spring 2024
Introduction to Information Science and Technology (SI100B), Spring 2020

PROFESSIONAL SERVICE

Reviewer

IEEE Transactions on Green Communications and Networking (TGCN), 2026
IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), 2023